

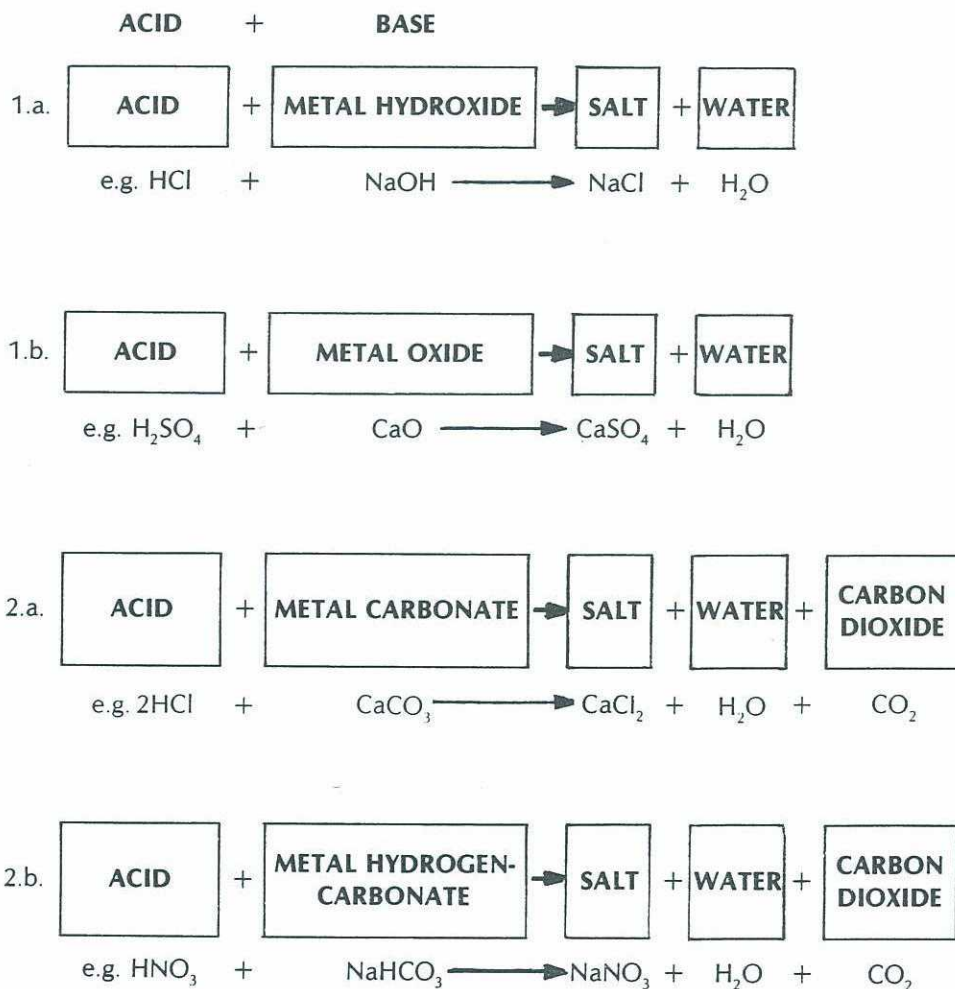
CHAPTER 7: General Reactions

Substances with similar chemical properties will react in similar ways and therefore many chemical reactions can be classified into general reaction types. Knowledge of general reactions enables products of specific reactions to be predicted.

The most common general reaction types will be treated here.

NEUTRALIZATION

Neutralization is the term used to describe the reaction between an *acid* and a *base*. The general equations for the many different forms of acids and bases are indicated here:



An **ACID** is a substance that produces hydrogen ions (H^+) in solution.

e.g. (i) Common acid: $\text{HCl(aq)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

(ii) Non-metal oxide: $\text{SO}_2(\text{g}) + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3(\text{aq}) \rightleftharpoons 2\text{H}^+(\text{aq}) + \text{SO}_3^{2-}(\text{aq})$
 $\text{SO}_3(\text{g}) + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4(\text{aq}) \rightleftharpoons 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
 $\text{CO}_2(\text{g}) + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons 2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 $\text{P}_4\text{O}_{10}(\text{g}) + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4(\text{aq}) \rightleftharpoons 12\text{H}^+(\text{aq}) + 4\text{PO}_4^{3-}(\text{aq})$



NOTE: 1. The use of $\rightleftharpoons$ indicates that the reaction has not gone to completion.

2. The acid in any of the general neutralization reactions above could be a non-metal oxide, in which case the salt formed is due to the acid produced when the non-metal oxide dissolves in water.

e.g. $\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$ (equation type (1.a.))

3. Sulfur dioxide is a major pollutant causing "acid rain".

The following acids should be memorised:

NAME OF ACID	FORMULA	USE
Hydrochloric acid	HCl	Cleans bricks and iron
Sulfuric acid	H_2SO_4	Car batteries, make fertilizers
Nitric acid	HNO_3	Make fertilizers, explosives
Phosphoric acid	H_3PO_4	Make fertilizers
Acetic acid	CH_3COOH	Vinegar
Carbonic acid	H_2CO_3	—
Sulfurous acid	H_2SO_3	—

Acids taste sour, feel rough and are corrosive.

A **BASE (or alkali)** is a substance that produces hydroxide ions (OH^-) in solution.

e.g. (i) Metal hydroxide: $\text{NaOH(aq)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$

(ii) Metal oxide: $\text{CaO(s)} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$

(iii) Metal carbonate: $\text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O} \rightleftharpoons 2\text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq}) + \text{HCO}_3^-(\text{aq})$

(iv) Metal hydrogencarbonate: $\text{KHCO}_3(\text{s}) + \text{H}_2\text{O} \rightleftharpoons \text{K}^+(\text{aq}) + \text{OH}^-(\text{aq}) + \text{H}_2\text{CO}_3(\text{aq})$

NOTE:1. Although they contain no metal ions, both ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, and ammonium hydrogencarbonate, NH_4HCO_3 , also behave as bases.

2. Ammonia solution is a common base. Its property of being a base can be seen in the following:



The following bases should be memorised:

COMMON NAME OF BASE	FORMULA	USE
Caustic soda	NaOH	Make soap, clean drains
Milk of magnesia	$\text{Mg}(\text{OH})_2$	Relieve an upset stomach
Ammonia	NH_3	Make fertilisers, explosives
Washing soda	Na_2CO_3	Make glass, soften water
Baking soda	NaHCO_3	Bake cakes
Quicklime	CaO	} Make lime mortar, plaster
Slaked lime	$\text{Ca}(\text{OH})_2$	

Bases taste bitter, feel slippery and are corrosive. They are used in household cleaners, dishwasher powder and oven cleaners to remove grease and fats.

EXERCISE 37: ACIDS AND BASES

Write the formula or name of each of the following substances and then classify it as an acid or a base:

- | | | |
|--------------------------------|----------------------------|----------------------|
| 1. Carbon dioxide | 5. Barium carbonate | 9. NH_3 |
| 2. Magnesium oxide | 6. KOH | 10. Nitrogen dioxide |
| 3. Li_2O | 7. H_3PO_4 | |
| 4. $\text{Ca}(\text{HCO}_3)_2$ | 8. Sulfur trioxide | |

A **SALT** is produced by the combination of the positive ion from the base and the negative ion from the acid.

For example, $\text{NaCl}(\text{aq})$ is the salt formed when $\text{HCl}(\text{aq})$ is reacted with $\text{NaOH}(\text{aq})$.

WATER is produced in neutralization through the combination of the hydrogen ion, $\text{H}^+(\text{aq})$, from the acid and the hydroxide ion, $\text{OH}^-(\text{aq})$, from the base.

EXAMPLE 21: Write the balanced equation for the reaction between dilute hydrochloric acid and copper(II) oxide solid.

ANSWER: By equation type (1.b.)



EXERCISE 38: NEUTRALIZATION

Write the balanced equations for the reactions produced when the following substances are mixed:

- Dilute sulfuric acid + potassium hydroxide solution.
- Dilute hydrochloric acid + calcium hydroxide.
- Dilute hydrochloric acid + magnesium oxide.
- Sulfur dioxide gas + sodium hydroxide solution.
- Calcium carbonate + dilute nitric acid.
- Acetic acid solution + iron(II) oxide.
- Aluminium hydrogencarbonate + sulfuric acid solution.
- Aluminium oxide + dilute hydrochloric acid.
- Calcium hydroxide + phosphoric acid solution.
- Potassium hydroxide solution + sulfur trioxide gas.
- Sulfurous acid solution + iron(III) hydrogencarbonate solution.
- Barium hydroxide solution + phosphoric acid solution.
- Nickel carbonate + dilute sulfuric acid.
- Tin(II) hydrogencarbonate + dilute hydrochloric acid.
- Dilute acetic acid + zinc hydroxide.
- Ammonia solution + dilute nitric acid.
- Carbonic acid + sodium hydroxide solution.
- Tetraphosphorus decaoxide + potassium hydroxide solution.
- Ammonium hydrogencarbonate solution + dilute hydrochloric acid.
- Dilute nitric acid + iron(III) hydroxide.

INDICATORS

An **INDICATOR** is a substance, often a plant dye, which has a characteristic but different colour in acidic and basic solutions.

INDICATOR	COLOUR IN ACID	COLOUR IN BASE
Litmus	Red	Blue
Methyl orange	Red	Yellow
Phenolphthalein	Colourless	Purple

EXERCISE 39: INDICATORS

State the colour of litmus, methyl orange and phenolphthalein in solutions of each of the substances listed in Exercise 37.

REACTION BETWEEN AN ACID AND A METAL



In this reaction the metal is displacing the hydrogen from the acid and therefore can be considered as a special example of displacement reactions (see p. 39). Therefore **ONLY METALS ABOVE HYDROGEN** on the Activity Series will obey this reaction (see inside back cover).

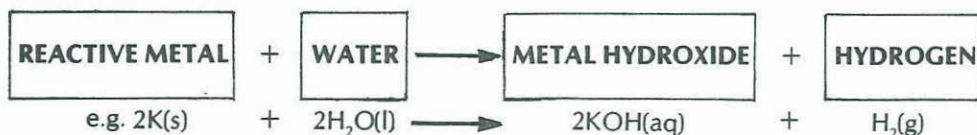
- NOTE: 1. Neither nitric acid nor concentrated sulfuric acid obey this equation.
2. When Fe reacts by this equation it produces Fe^{2+} ions.
3. When Sn reacts by this equation it produces Sn^{2+} ions.
4. When Pb reacts by this equation it produces Pb^{2+} ions.

EXERCISE 40: ACID-METAL REACTIONS

Write balanced equations for the reactions produced when the following substances are mixed. If a reaction does not occur, indicate by writing "no reaction".

1. Zinc + dilute hydrochloric acid.
2. Magnesium + dilute sulfuric acid solution.
3. Aluminium + hydrochloric acid.
4. Phosphoric acid + calcium.
5. Copper + dilute hydrochloric acid.
6. Iron + dilute sulfuric acid.
7. Aluminium + acetic acid solution.
8. Dilute sulfuric acid + silver.
9. Dilute sulfuric acid + lead.
10. Nickel + dilute hydrochloric acid.

REACTION BETWEEN A REACTIVE METAL AND WATER



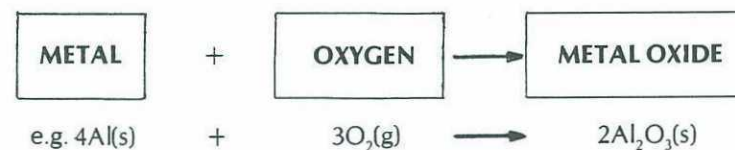
This reaction occurs *only for the elements at the top of the Activity Series (potassium, sodium and calcium)*. However, if the water is converted to steam, then other metals (e.g. Mg, Al, Fe, Zn), towards the top of the Activity Series (see inside back cover) can be made to react in this way to form a metal oxide and hydrogen.

EXERCISE 41: WATER-METAL REACTIONS

Write the balanced equation for the reaction produced when the following substances are mixed:

1. Sodium + water.
2. Calcium + water.

REACTION BETWEEN A METAL AND OXYGEN



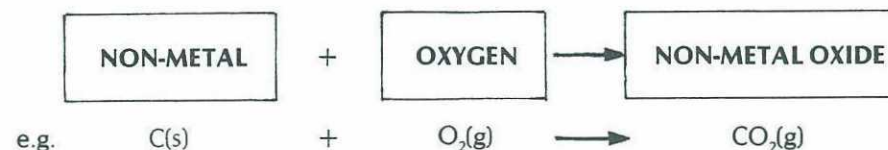
This reaction is often referred to as **OXIDATION**, although if it proceeds very rapidly it is said to be **COMBUSTION** or **burning**. The ability to react with oxygen will vary according to the Activity Series (see inside back cover).

EXERCISE 42: OXYGEN-METAL REACTIONS

Write the balanced equations for the following reactions.

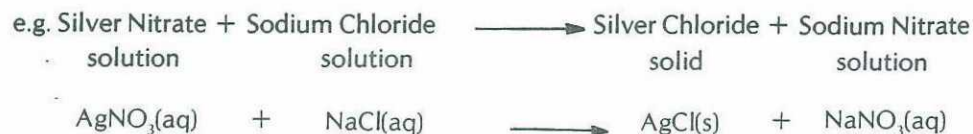
1. Reaction of calcium and oxygen.
2. Oxidation of zinc by oxygen.
3. Burning of magnesium.
4. Production of iron(III) oxide from the oxidation of iron.
5. Production of lead(II) oxide from the oxidation of lead.
6. Formation of copper(II) oxide from the oxidation of copper.
7. Oxidation of nickel by oxygen.
8. Production of iron(II) oxide from the oxidation of iron.
9. Formation of mercury(II) oxide from the oxidation of mercury.
10. Oxidation of tin to tin(II) oxide by oxygen.

REACTION BETWEEN A NON-METAL AND OXYGEN



PRECIPITATION REACTIONS

When two different salt solutions are mixed, a precipitate will occur if, by combination of a positive ion and a negative ion, a substance of low solubility is produced.



To predict products of precipitation reactions a knowledge of *solubility rules* is required. The more common rules are indicated here.

SOLUBILITY OF IONIC COMPOUNDS IN WATER

Soluble Compounds	Exceptions
Halides $\left[\begin{array}{l} \text{Chlorides (Cl}^-) \\ \text{Bromides (Br}^-) \\ \text{Iodides (I}^-) \end{array} \right]$	$\text{Ag}^+, \text{Pb}^{2+}, \text{Hg}_2^{2+}, \text{Cu}^+$
Nitrates (NO_3^-)	none
Acetates (CH_3COO^-)	none
Sulfates (SO_4^{2-})	$\text{Ca}^{2+}, \text{Ba}^{2+}, \text{Pb}^{2+}$
Low Solubility Compounds	Exceptions
Carbonates (CO_3^{2-})	$\text{Li}^+, \text{Na}^+, \text{K}^+, \text{H}^+, \text{NH}_4^+$
Hydroxides (OH^-)	$\text{Li}^+, \text{Na}^+, \text{K}^+, \text{H}^+, \text{NH}_4^+, \text{Ba}^{2+}$

NOTE: From this table it can be concluded that:

all $\left[\begin{array}{l} \text{lithium} \\ \text{sodium} \\ \text{potassium} \\ \text{ammonium} \end{array} \right]$ compounds are soluble.

EXERCISE 46: PRECIPITATION REACTIONS

Write balanced equations for any of the following mixtures where a precipitate is produced. Be sure to use subscripts to indicate the form of the substance. If a reaction does not occur indicate by writing "no reaction".

1. Calcium chloride solution + sodium carbonate solution.
2. Barium hydroxide solution + dilute sulfuric acid.
3. Sodium nitrate solution + sodium chloride solution.
4. Potassium nitrate solution + lead(II) acetate solution.
5. Lead(II) nitrate solution + lead(II) acetate solution.
6. Sodium hydroxide solution + calcium chloride solution.
7. Ammonium carbonate solution + aluminium nitrate solution.
8. Magnesium nitrate solution + sodium carbonate solution.
9. Copper(II) sulfate solution + aluminium chloride solution.
10. Iron(III) sulfate solution + potassium carbonate solution.

EXERCISE 47: MISCELLANEOUS I

Write balanced equations for the following reactions. If no reaction occurs indicate by writing "no reaction". Use subscripts to indicate the phase of the substance.

1. Dilute sulfuric acid is poured over solid iron(II) hydrogencarbonate.
2. Burning of carbon in a plentiful supply of oxygen.
3. Heating of potassium carbonate.
4. Solid magnesium oxide is added to dilute nitric acid.
5. Ammonia solution is added to sodium hydroxide solution.
6. A piece of zinc is heated strongly in an atmosphere of oxygen.
7. Acetic acid solution is poured onto solid calcium oxide.
8. A copper bangle is dropped into silver nitrate solution.
9. Solid calcium hydrogencarbonate is heated with a bunsen burner.
10. Dilute sulfuric acid is poured onto solid lead(II) carbonate.
11. The "lime-water" test (carbon dioxide is bubbled through calcium hydroxide solution).
12. The "pop-test" for hydrogen (hydrogen is burnt in air).
13. The mixing of iron(II) chloride solution and sodium hydroxide solution.
14. Mercury is poured into dilute hydrochloric acid.
15. Lead(II) nitrate solution and potassium iodide solutions are mixed.
16. Neutralization of tin(II) hydroxide with dilute sulfuric acid.
17. Oxidation of calcium in air.
18. Heating of iron(II) carbonate with a bunsen burner.
19. Steam is blown over a piece of magnesium.
20. Copper(II) oxide solid is added to dilute hydrochloric acid.